

Jagger C. Walraven

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Gameplay Programmer and Developer with experience gained from 2 unique internships and over 5 Unity projects as Lead Programmer.

SUMMARY

Capable developer responsible for building foundational gameplay systems including player controllers, AI navigation and decision trees, procedural level generators, projectile object pools, and event-based UI receivers. Comfortable with many key Unity libraries such as Cinemachine, Input System, UI, Events, and Splines. Successful in creating experiences spanning 7 different genres, 3 of which are award-winning.

SKILLS & TOOLS

Unity Engine ■ C# ■ Data Structures ■ Design Patterns ■ Agile Methodologies ■ Mobile Games ■ Atlassian Products ■ Documentation ■ Git ■ GitHub ■ Visual Studio ■ Visual Studio Code ■ Vector Algebra ■ Trigonometry ■ Matrices ■ Discrete Math ■ Leadership ■ Time Management

RECENT EXPERIENCE

Gameplay Programmer, Lionheart Studios LLC

Apr 2026 - Present

- Gameplay programmer for an unannounced game with a planned release on Steam.

Gameplay Programmer, Games For Love

Feb 2026 - Apr 2026

- Led AI development in a team of 8 programmers**, utilizing state machines and object pooling to ensure high performance and scalability.
- Built custom inspector tool** allowing designers to tweak AI spawner values, **saving around 3 hours** of quality assurance and testing.
- Corresponded with the engineering team daily** and **with the entire team weekly** through 15-minute scrum meetings.

Immersive Simulation Developer, USF College of Health Sciences

Oct 2024 - Oct 2025

- Created 2 medical imaging and training scenarios** for the College's immersive laboratory to be **used in the nursing curriculum**, using Unity and proprietary software.

EDUCATION

University of Saint Francis

2022 - 2026

Bachelor of Science in Game and Application Development

Cumulative GPA: 3.99

- Concentration in Programming
- Minors in Computer Science and Film
- President of Game Development Club 2024 - 2025
- Captain of Super Smash Bros. Ultimate eSports Team 2024 - 2026

PROJECTS

Uphill Battle, Lead Programmer

Unity 3D, 2025

- Utilized knowledge on design patterns to **construct a custom procedural level generator**, complete with **9 tunable settings**, seed input/output, and texture randomizer.
- Designed and **implemented node-based AI decision tree**, allowing enemies to **consider 13 scenarios** before choosing one of **8 outcomes**.
- Connected backend gameplay variables and frontend UI elements **using modular, event-driven subject-observer pattern**.

30 Minutes or it's Free, Lead Programmer

Unity 3D, 2025

- Developed **responsive, parkour-oriented player controller**, complete with **save system that tracked 4 key values** that allowed for asynchronous play.
- Supplied level designers with a **custom NPC placement tool and documentation** to expedite work, allowing for **7 extra levels** than planned.
- The project received USF's Department of Communication, Animation, and Technology **Best Game Demo 2026 award**.

Fortune Queller, Lead Programmer

Unity 3D, 2025

- Constructed an **efficient card-based gameplay system** using maps and queues, **complete with 11 distinct cards** that altered the player's actions and stats.
- The project received USF's Department of Communication, Animation, and Technology **Best Game Demo 2025 award**.

Diggy Dog, Lead Programmer

Unity 3D, 2024

- Employed knowledge of **trigonometric functions and Cinemachine** to create a radial camera for accurate player control.
- Implemented **custom interactables system for 4 different items**, including push and pull blocks, ladders, key doors, and holes.

Toilet Time Traveler

Unity 2D, 2024

- Led development of **3 unique microgames**, including programming, spritework, and sound design.
- Used **Input System library for touch interpretation**, allowing the game to target **Android and iOS** devices.
- The project received USF's Department of Communication, Animation, and Technology **Best Game Demo 2024 award**.